ECON 381: Homework 4

Fall 2026

Due: Thursday, November 12, 2026, at 11:59 p.m. Eastern Time
Submit electronically through the Canvas **Assignments** section.

Instructions and submission guidelines

1. Complete **all ten questions** for **100 points**: 40 for the empirical exercise, 40 for five short-answer questions, and 20 for four true-or-false questions. The assignment covers Chapters 6 and 7 (Handouts 7 and 8).

2. You may submit individually or in a group of two, as allowed by the syllabus. A pair should make **one group submission** and list both names at the top of the homework PDF and in comments at the top of the do-file. Group members receive the same grade.

3. Submit **all three** of the following on Canvas:

   - a **single PDF of handwritten solutions** for the short-answer and true-or-false questions;
   - a **Stata do-file** containing your code for E7.2; and
   - a **Stata log file or PDF of printed Stata results** for E7.2.

   Write your E7.2 answers by hand in the homework PDF or as comments in the do-file. Label every exercise and subpart clearly. **No credit** will be given for E7.2 if either the do-file or the log/results file is missing.

4. Scan your handwritten solutions or export them from a tablet. Check that **every page** is present and legible before submitting. Illegible submissions will receive **no credit.**

5. Use the same filename base for all submitted files.

   Individual: `ECON381_HW4_LastName`
   Pair: `ECON381_HW4_LastName1_LastName2`

   Append `.pdf` for the handwritten solutions, `.do` for the do-file, and either `.log` for the Stata log or `_Output.pdf` for printed Stata results.

6. Late work is accepted according to the syllabus policy.

Time submitted after the deadline	Percentage of earned score retained
At most 3 hours	80%
More than 3 and at most 6 hours	60%
More than 6 and at most 9 hours	40%
More than 9 and at most 12 hours	20%
More than 12 hours	0%

   Plan ahead when you anticipate a conflict. Illness, university events, and other documented circumstances do not ordinarily excuse a late submission.

7. Any use of generative AI must follow the syllabus disclosure and attribution policy. If you use such a tool, include the required acknowledgment naming the tool and provider and explaining the purpose for which it was used.

1 Empirical Exercise (40 points)

E7.2. Height, education, and earnings

Adapted from James H. Stock and Mark W. Watson, Introduction to Econometrics, 4th ed. (Pearson), Empirical Exercise E7.2, pp. 231–232.

In Homework 3, you found a relatively large, statistically significant association between height and earnings. Case and Paxson (2008) suggest that poor nutrition and other harmful conditions before birth and in early childhood can impair both physical and cognitive development. Cognitive ability can in turn affect earnings later in life.

(a) Suppose this mechanism is correct. Explain why omitting cognitive ability biases the OLS slope in the regression of *Earnings* on *Height*. Would the estimated slope tend to be too large or too small? *Hint.* Review the omitted variable bias formula in Equation (6.1) of the textbook or in Handout 7. **[2 points]**

The dataset records years of education, which may help control for cognitive ability if students with higher ability stay in school longer. Use `educ` to create the following four indicators. Set each indicator to 1 when its condition holds and to 0 otherwise. **[4 points]**

Indicator	Education level	Condition
`LT_HS`	Less than a high school diploma	`educ` < 12
`HS`	High school diploma	`educ` $= 12$
`Some_Col`	Some college	$12 <$ `educ` < 16
`College`	Bachelor's degree or higher	`educ` ≥ 16

(b) Using data for women only, regress (1) *Earnings* on *Height* and (2) *Earnings* on *Height*, `LT_HS`, `HS`, and `Some_Col`. Use heteroskedasticity-robust standard errors. **[15 points]**

(i) Compare the estimated coefficient on *Height* across the two regressions. Is the change large? Is it consistent with the cognitive ability explanation? Explain. **[7 points]**

(ii) Why does regression (2) omit `College`? **[1 point]**

(iii) Test the joint null hypothesis that the coefficients on the included education indicators are all zero. **[5 points]**

(iv) Interpret the three education coefficients. Why are they all negative, with `LT_HS` the most negative and `Some_Col` the least negative? **[2 points]**

(c) Repeat part (b) using data for men with the same subpart weights. **[15 points]**

Data and Stata instructions

- Download `Earnings_and_Height.zip` from **Textbook Dataset** on Canvas. Unzip it to get `Earnings_and_Height.dta` and `Earnings_and_Height_Description.pdf`.
- Your do-file must run from beginning to end without errors. Label subparts with comments and place any written answers after the relevant commands. E7.2 is graded on answers (16 points), code (20 points), and log/results and documentation (4 points). Record a Stata log or print to PDF from `File` → `Print` → `Results`.
- For Stata help, email the TA Bowen Jiang at bjiang12@vols.utk.edu.

2 Short Answer Questions (40 points)

1. What is the *dummy variable trap*? Explain why it creates perfect multicollinearity in a regression. How can we avoid it? **[4 points]**

2. What is the difference between a *treatment variable* and a *control variable*? How should researchers select control variables? **[4 points]**

3. Consider the regression

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Education}_i + u_i,$$

where the sample consists of individuals ages 30 to 35. The variable Earnings_i measures annual earnings, and Education_i measures years of education.

Consider two omitted variables, years of work experience and cognitive ability. For each of these two omitted variables separately, would omitting it tend to make the estimated effect of education on earnings too large or too small? Explain your reasoning. **[6 points]**

4. The table below reports Stata results for a regression of average district test scores (TestScr_i) on the student–teacher ratio (STR_i), computers per student (Comp_Stu_i), and average district income (AvgInc_i). Computers per student could be regarded as a measure of school expenditure on students. **[12 points]**

Table 1. Stata results for the district test-score regression

Regressor	Coefficient	Robust std. error	t	p	95% confidence interval Lower	Upper
`str`	−0.2882447	0.3637797	−0.79	0.429	−1.00332	0.4268308
`comp_stu`	37.93661	10.19393	3.72	0.000	17.89857	57.97465
`avginc`	1.794593	0.1157663	15.50	0.000	1.567034	2.022153
`_cons`	627.1742	7.958022	78.81	0.000	611.5312	642.8171

$n = 420$, $F(3, 416) = 88.13$, Prob $> F = 0.0000$, $R^2 = 0.5264$, Root MSE $= 13.16$.

(a) Fill in the blanks, rounding the coefficients, parenthesized values, and R^2 to two decimal places. What do the parenthesized values represent? **[3 points]**

$$\widehat{\text{TestScr}}_i = \underset{(____)}{____} + \underset{(____)}{____}\,\text{STR}_i + \underset{(____)}{____}\,\text{Comp_Stu}_i + \underset{(____)}{____}\,\text{AvgInc}_i,$$

$$R^2 = ____, \qquad n = ____.$$

(b) Interpret the estimated coefficient on STR_i. **[2 points]**

(c) The regression controls for AvgInc_i in place of the percentage of English learners. Is AvgInc_i a good control for estimating the causal effect of STR_i on test scores? Explain. **[2 points]**

(d) Is the coefficient on STR_i statistically significant at the 10% level? What about the coefficient on Comp_Stu_i? **[2 points]**

(e) Can you test whether at least one of the coefficients on STR_i and Comp_Stu_i is nonzero using only the individual t-statistics? Explain. Which test would you use for this hypothesis, and what Stata command would you run? **[3 points]**

5. **Exercise 7.2. Average hourly earnings. [14 points]**

Adapted from Stock and Watson, Introduction to Econometrics, 4th ed., pp. 228–229. Parts (c)–(d) extend the textbook exercise.

Table 2 uses 2015 Current Population Survey data. The outcome variable is average hourly earnings in dollars. *College* equals 1 for college graduates and 0 for high school graduates. *Female* equals 1 for women and 0 for men. *Age* is measured in years. *Northeast*, *Midwest*, *South*, and *West* each equal 1 if the worker is in that region and 0 otherwise.

Table 2. Regressions of average hourly earnings

Regressor	(1)	(2)	(3)
College	10.47	10.44	10.42
	(0.29)	(0.29)	(0.29)
Female	−4.69	−4.56	−4.57
	(0.29)	(0.29)	(0.29)
Age		0.61	0.61
		(0.05)	(0.05)
Northeast			0.74
			(0.47)
Midwest			−1.54
			(0.40)
South			−0.44
			(0.37)
Intercept	18.15	0.11	0.33
	(0.19)	(1.46)	(1.47)
F-statistic for zero regional effects			**9.32**
SER	12.15	12.03	12.01
R^2	0.165	0.182	0.185
n	7,178	7,178	7,178

Standard errors are in parentheses. Blank entries indicate excluded regressors or unreported statistics.

For parts (a) and (b), report the estimated gap and show how you reached your test decision.

(a) Using column (1), test whether the college-minus-high-school earnings difference is statistically significant at the 5% level. Construct a 95% confidence interval for the difference. **[4 points]**

(b) Using column (1), test whether the **male-minus-female** earnings difference is statistically significant at the 5% level. Construct a 95% confidence interval for the difference in that order. **[4 points]**

(c) What would happen if the researchers added *West* to column (3) while keeping the intercept and all the other regressors? In column (3) as reported, how do you interpret the coefficient on *Northeast*? **[2 points]**

(d) Write the joint null hypothesis for the regional-effects F-test in column (3). Which large-sample reference distribution from Handout 8 would you use to obtain the critical value for the F-test? Say the 5% critical value is 2.60. Does the reported F-statistic provide evidence of regional differences in hourly earnings after controlling for the other regressors? **[4 points]**

3 True/False Questions (20 points)

Each statement is worth 5 points. Determine whether it is true or false and briefly justify your answer.

1. We should always try to maximize R^2 or adjusted R^2 if our goal is to estimate a causal effect using regression.

2. Under the conditional mean independence assumption

$$E(u_i \mid X_i, W_i) = E(u_i \mid W_i),$$

   the coefficient $\hat{\beta}_2$ in the regression $\widehat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 W_i$ estimates the causal effect of the control variable W_i.

3. Adjusted R^2 can decrease when more regressors are added, whereas R^2 never decreases when more regressors are added to a regression with an intercept on the same sample.

4. If two regressors are individually (in)significant based on their own t-statistics, then they must also be jointly (in)significant in an F-test.